Supplementary Material: Benchmark Definitions, Execution Protocol, and Uncertainty

This supplement to *Black-Box Configuration Search for Generalized Disjunctive Programming* provides benchmark definitions, execution and acceptance rules, model sizes, seed-level uncertainty, and the numerical evidence underlying the algebraic bound-consistency checks. The analytical functions are adapted from the Quantum Hamiltonian Descent collection; each instance is defined by its implemented expression and domain rather than its conventional name (SECQUOIA, 2026b; Wu Group, n.d.).

S1 Coordinate domains and normalization

For each row of Table S1, both endpoints are included and the physical coordinates are

$$a_i = \ell_x + \frac{i}{30}(u_x - \ell_x), \qquad b_j = \ell_y + \frac{j}{30}(u_y - \ell_y), \qquad i, j \in \{0, \ldots, 30\}. \tag{S1}$$

The raw objective f is defined below. The minimized objective is

$$F(x, y) = \frac{f(x, y) - L}{U - L}, \qquad L = \min_{i,j} f(a_i, b_j), \quad U = \max_{i,j} f(a_i, b_j). \tag{S2}$$

Thus L and U are obtained by exhaustive preprocessing of the 961 lattice points, not from the continuous optimum. All trigonometric arguments are in radians; $e = \exp(1)$ and π is the usual circular constant.

Table S1: Analytical Grid-GDP benchmark domains. All intervals include both endpoints and have equally spaced coordinates. Each row defines one independent problem with 961 configurations and 62 Boolean indicators. Historical function expressions and domains are preserved (SECQUOIA, 2026b; Wu Group, n.d.).

Function	x domain	y domain	Grid
Ackley	$[-5, 5]$	$[-5, 5]$	31×31
Alpine 1	$[0, 10]$	$[0, 10]$	31×31
Alpine 2	$[0, 1000]$	$[0, 1000]$	31×31
Bohachevsky 2	$[-100, 100]$	$[-100, 100]$	31×31
Three-hump camel	$[-5, 5]$	$[-5, 5]$	31×31
Csendes (squared variant)	$[-1, 1]$	$[-1, 1]$	31×31
Deflected corrugated spring	$[0, 10]$	$[0, 10]$	31×31
Drop-Wave	$[-5.12, 5.12]$	$[-5.12, 5.12]$	31×31
Easom	$[-10, 10]$	$[-10, 10]$	31×31
Griewank	$[-600, 600]$	$[-600, 600]$	31×31
Hölder table	$[-10, 10]$	$[-10, 10]$	31×31
Hosaki	$[0, 5]$	$[0, 6]$	31×31
Lévy	$[-10, 10]$	$[-10, 10]$	31×31
Lévy 13	$[-10, 10]$	$[-10, 10]$	31×31
Michalewicz	$[0, \pi]$	$[0, \pi]$	31×31
Rastrigin	$[-5.12, 5.12]$	$[-5.12, 5.12]$	31×31
Rosenbrock	$[-2, 2]$	$[-1, 3]$	31×31
Shubert (historical variant)	$[0, 10]$	$[0, 10]$	31×31
Styblinski–Tang	$[-5, 5]$	$[-5, 5]$	31×31
Sum of squares	$[-10, 10]$	$[-10, 10]$	31×31
Xin-She Yang 3 (historical variant)	$[-20, 20]$	$[-20, 20]$	31×31

S2 Implemented analytical objectives

Each expression below defines a benchmark on the corresponding domain in Table S1, with all coefficients fixed. The definitions follow `gdp_models/grid.py`; differences from the historical `qhdopt_gdp.py` are noted where relevant (SECQUOIA, 2026b, n.d.).

1. Ackley

$$\begin{aligned} f(x,y) = & -20\exp\left(-0.2\sqrt{\frac{x^2+y^2}{2}}\right) \\ & -\exp\left(\frac{\cos(2\pi x)+\cos(2\pi y)}{2}\right)+20+e. \end{aligned} \tag{S3}$$

2. Alpine 1

$$f(x,y) = |x\sin x + 0.1x| + |y\sin y + 0.1y|. \tag{S4}$$

3. Alpine 2

$$f(x,y) = -\sqrt{xy}\,\sin x \sin y. \tag{S5}$$

The negative sign and the domain $[0,1000]^2$ are retained.

4. Bohachevsky 2

$$f(x,y) = x^2 + 2y^2 - 0.3\cos(3\pi x)\cos(4\pi y) + 0.3. \tag{S6}$$

5. Three-hump camel

$$f(x,y) = 2x^2 - 1.05x^4 + \frac{x^6}{6} + xy + y^2. \tag{S7}$$

6. Csendes (squared variant)

$$f(x,y) = c(x) + c(y), \qquad c(t) = \begin{cases} t^6\left[2+\sin(1/t)\right]^2, & t \neq 0, \\ 0, & t = 0. \end{cases} \tag{S8}$$

The square applies to the entire oscillatory factor. The current GDP uses branch-fixed auxiliaries $r_t = 1/t$ for $t \neq 0$ and $r_t = 0$ for $t = 0$, evaluating $t^6[2+\sin(r_t)]^2$ without division by zero.

The archived `qhdopt_gdp.py` instead uses $c_\varepsilon(t) = t^6[2+\sin(1/(t+\varepsilon))]^2$ with $\varepsilon = 10^{-12}$ (SECQUOIA, n.d.). This also gives zero at $t = 0$, but is not algebraically identical to Equation (S8) at nonzero coordinates. The current implementation uses the exact zero extension, with no epsilon shift; its normalized values agree with the archived grid to the stated 10^{-12} absolute verification tolerance.

7. Deflected corrugated spring

With $r = \sqrt{(x-5)^2 + (y-5)^2}$,

$$f(x,y) = 0.1r^2 - \cos(5r). \tag{S9}$$

8. Drop-Wave

$$f(x,y) = -\frac{1 + \cos\left(12\sqrt{x^2+y^2}\right)}{0.5(x^2+y^2)+2}. \tag{S10}$$

9. Easom

$$f(x,y) = -\cos x \cos y \, \exp\left(-[(x-\pi)^2 + (y-\pi)^2]\right). \tag{S11}$$

10. Griewank

$$f(x,y) = \frac{x^2+y^2}{4000} - \cos x \cos\left(\frac{y}{\sqrt{2}}\right) + 1. \tag{S12}$$

11. Hölder table

$$f(x,y) = -\left|\sin x \cos y \, \exp\left(\left|1 - \frac{\sqrt{x^2+y^2}}{\pi}\right|\right)\right|. \tag{S13}$$

12. Hosaki

$$f(x,y) = \left(1 - 8x + 7x^2 - \frac{7}{3}x^3 + \frac{1}{4}x^4\right) y^2 \exp(-y). \tag{S14}$$

13. Lévy

Define $w_x = 1 + (x-1)/4$ and $w_y = 1 + (y-1)/4$. Then

$$\begin{aligned} f(x,y) = \; & \sin^2(\pi w_x) \\ & + (w_x - 1)^2\big[1 + 10\sin^2(\pi w_x + 1)\big] \\ & + (w_y - 1)^2\big[1 + \sin^2(2\pi w_y)\big]. \end{aligned} \tag{S15}$$

The middle sine argument is $\pi w_x + 1$.

14. Lévy 13

$$\begin{aligned} f(x,y) = \; & \sin^2(3\pi x) + (x-1)^2\big[1 + \sin^2(3\pi y)\big] \\ & + (y-1)^2\big[1 + \sin^2(2\pi y)\big]. \end{aligned} \tag{S16}$$

15. Michalewicz

$$f(x,y) = -\sin x \, \sin^{20}\left(\frac{x^2}{\pi}\right) - \sin y \, \sin^{20}\left(\frac{2y^2}{\pi}\right). \tag{S17}$$

The fixed exponent is 20 (equivalently $2m$ with $m = 10$).

16. Rastrigin

$$f(x,y) = x^2 - 10\cos(2\pi x) + y^2 - 10\cos(2\pi y) + 20. \tag{S18}$$

17. Rosenbrock

$$f(x,y) = (1-x)^2 + 100(y - x^2)^2. \tag{S19}$$

On the stated grid in $[-2,2] \times [-1,3]$, the exact normalization constants are $L = 13/2025$ and $U = 2509$, as in the main manuscript.

18. Shubert (historical variant)

$$\begin{aligned} f(x,y) = \; & \big[\cos(2x+1) + 2\cos(3x+2) + 3\cos(4x+3)\big] \\ & \times \big[\cos(2y+1) + \cos(y+2)\big]. \end{aligned} \tag{S20}$$

This is the implemented three-term by two-term product. In particular, the final term is $\cos(y+2)$ with coefficient one.

19. Styblinski–Tang

$$f(x,y) = \frac{1}{78}\,\frac{1}{2}\left(x^4 - 16x^2 + 5x + y^4 - 16y^2 + 5y\right). \tag{S21}$$

The historical factor $1/78$ is retained in the raw objective, even though positive scaling cancels in the normalization of Equation (S2).

20. Sum of squares

$$f(x,y) = x^2 + 2y^2. \tag{S22}$$

21. Xin-She Yang 3 (historical variant)

$$f(x,y) = \exp\!\left(-\left[\left(\frac{x}{15}\right)^6 + y^6\right]\right) + 2\exp\!\left(-(x^2+y^2)\right)\cos^2 x \cos^2 y. \tag{S23}$$

The exponential–cosine term has a positive coefficient $+2$. Only x is divided by 15 in the first exponential; y^6 is unscaled. Floating-point exponential underflow can produce stored zero values; these do not change the analytical expression.

S3 Source verification

All 21 definitions and all domain bounds were verified against `gdp_models/grid.py` at repository commit `cb70771b` and `qhdopt_gdp.py` at historical commit `11ba8728`; the full revisions are pinned by the source links below. The current implementation's SHA-256 digest matches that recorded in `results/grid_gdp_20261009/raw/run.json`. Exhaustive evaluation of the implemented analytical GDP objectives at all $21 \times 961 = 20{,}181$ lattice points agrees with the archived `benchmark_grids/*.csv` normalized values within absolute tolerance 10^{-12}. The target sets at $F \le 10^{-5}$ also agree.

S4 Executable protocol and implementation settings

The companion protocol files and `grid_execution_records.json` preserve the historical execution settings, with `protocol_rbfopt.json` referring specifically to the solver variant. The additional archive supplies the matched initial coordinates, seeds, budgets and actual settings for the primary RBFOpt results (Section S10).

For each process condition and seed (0–9), all methods share the same three integer Latin-hypercube starts. Python and NumPy use the job seed, and RBFOpt also receives `rand_seed`. BAM exposes no separate native random-seed option: its process wrapper seeds Python/NumPy, while the Grid seed identifies the common initial design. FMQA-SA uses three NumPy `SeedSequence(seed)` streams for FM initialization, annealing, and uniform unseen-point fallback; each training or annealing invocation draws a seed from its stream. PyTorch uses deterministic CPU operations and one thread.

The budget charges every objective callback, including initialization, repeats, and unsuccessful configurations; internal cache accesses without a callback are not charged. Repeated callbacks reuse the stored response and point cost. FMQA-SA excludes evaluated candidates and uses a uniformly selected unseen point when no annealed candidate is eligible; out-of-bounds proposals are discarded before evaluation. Logical and NLP feasibility are determined by the recorded oracle response.

Process DSDA evaluates all three starts and begins one descent from the best accepted point; if none is accepted, it fails without reseeding. Other methods retain failed observations under their declared non-finite/penalty handling. Grid DSDA ranks the initial points and attempts successive descents until reaching the budget or known reference target. It enumerates the full L_∞ neighborhood as the lexicographic Cartesian product of $(-1, 0, 1)$, excluding the zero direction and out-of-bounds points; ties select the first point, and improvement must exceed 10^{-12}.

Primary RBFOpt 4.3.0 uses MSRSM, `rbf=auto`, `global_search_method=genetic`, integer variables, and `do_init_strategy=False`. The recorded defaults are `ga_base_population_size=400` and `ga_num_generations=20`; only the global-search option differs from the retained solver settings. RBFOpt 4.3.0 uses an effective population of $400 + \lfloor 20n/5 \rfloor$, where n is the dimension passed to the genetic-search routine, giving 408 and 412 for two and three search variables. Recorded defaults include `max_iterations=1000`, `max_cycles=1000`, `max_stalled_iterations=100`, `max_clock_time=1e30`, `target_objval=-1e10`, and `eps_opt=0.01`. Bonmin solves surrogate MINLPs, and the configured IPOPT executable supports surrogate NLPs; these are distinct from GAMS/IPOPT process evaluations. BAM 5.0.0 receives `x0=None`, all-integer variables, the three evaluated starts, and `max_evals`=B-3, with other options at package defaults; its local models use ALAMO/BARON. Neither RBFOpt nor BAM receives the benchmark reference as a stopping target.

In the process runs, BAM returns callback B normally and stops at its native limit; the other wrappers record that callback before stopping (Section S4.3). FMQA-SA stops at B or its 1,800 s method limit, checked between operations and through training/sampling callbacks, without reference-target stopping. All process workers and the Grid FMQA worker have a 2,100 s watchdog. The Grid RBFOpt/BAM trace watchdogs are disabled (`max_run_seconds=0`); archived Grid DSDA uses budget, target, or local-search termination without an additional method-clock limit.

S4.1 Versions and independently timed process evaluations

The recorded environments include Python 3.11.16, RBFOpt 4.3.0, BAM 5.0.0, Pyomo 6.10.0, GAMS 52.2, BARON 25.11.17, and HiGHS 1.12.0. The process DSDA/RBFOpt/BAM environment records NumPy 2.4.6 and SciPy 1.17.1; the FMQA environment records NumPy 2.2.6, SciPy 1.15.3, PyTorch 2.8.0+cpu, dimod 0.12.21, and dwave-samplers 1.6.0. The annealer is the classical CPU implementation `dwave.samplers.SimulatedAnnealingSampler` (D-Wave Systems, n.d.). Executable banners identify Bonmin 1.8.9 with Cbc 2.10.12 and Ipopt 3.14.19, and standalone Ipopt 3.14.19, on Linux x86_64 with ASL 20231111. These auxiliary-solver identifiers are in `version_identification.json`; per-job binary hashes are in the protocol files. The `bam-solver` 5.0.0 engine identifies version 2026.9.10, build LNX-int-64, timestamp 2026-09-10 17:59:06. Its internal ALAMO and BARON versions are unspecified; the recorded GAMS/BARON 25.11.17 version applies only to that separate interface.

The fixed-configuration process evaluations use GAMS/IPOPT with MUMPS, solver tolerance 10^{-8} and a 60 s solver limit. The acceptance rule in `gdp_models/evaluation.py` requires all of the following: a finite objective; termination in {`optimal`, `locallyOptimal`, `globallyOptimal`}; solver status other than `error` or `aborted`; GAMS `SOLVESTAT`=1 and `MODELSTAT`$\in \{1, 2\}$; and a finite maximum primal violation no greater than 10^{-5}. The 10^{-8} solver tolerance is distinct from this acceptance threshold. Time-limit, iteration-limit, evaluation-limit, and error outcomes are not accepted, even if a finite feasible candidate is available.

The primal check takes the largest positive absolute constraint-bound or variable-bound violation, with zero for satisfied bounds and no row-wise normalization. Constraint rows are collected after fixing the disjunctions and before preprocessing, so rows subsequently deactivated by presolve are also checked. Bounds of initialized active variables are checked on the prepared NLP. An undefined or non-finite constraint body, a non-finite initialized variable, or a non-finite objective precludes acceptance. Uninitialized unused variables are skipped; the objective and all retained constraint expressions are still evaluated. Preprocessing rejection or worker failure supplies no accepted objective.

CSTR5–25 start from constructor defaults and then apply `cstr_box_ipopt.py:initialize_feasible` after preprocessing. This sets mass-balance-consistent values with zero recycle and checks the initial primal violation at 10^{-8} without changing constraints or bounds. CSTR30 retains constructor defaults. Column and Small Batch load `gdp/dsda/column_17_initialization.json` and `gdp/dsda/small_batch_initialization.json`, respectively, from the recorded `dsda-gdp` source revision, using its `gdp/dsda/model_serializer.py`. Dynamic clones the unsolved discretized template; it has no saved initialization vector. These are cold starts, with no transfer of an NLP solution between configurations. The companion `process_nlp_acceptance.json` identifies the acceptance code, initialization artifacts, hashes, and provenance. Objectives, acceptance status, and NLP point costs are linked to the same recorded attempt by model, CSTR size, and coordinate, without successful-retry selection or cross-size time substitution. Separately measured structural-rejection costs cover 435 CSTR30, 17 Column, and 35 Dynamic coordinates without invoking an NLP. Each stored cost is reused on revisits, so reconstructed costs do not capture repeat-measurement timing variability; Dynamic template preparation is added once per run.

S4.2 Observed proposal and initialization behavior

The Grid BAM results at $B = 10$ contain 111 hits before surrogate-guided selection: 11 in the common three-point initialization and 100 at two predetermined Sobol points (10 at a corner and 90 at the center). This is evidence about the configured initialization on these grids, not an isolated estimate of adaptive surrogate benefit. Table S2 pools FMQA-SA proposals across tested budgets and ten seeds per condition, including all six CSTR sizes. Counts exclude the shared initialization and include terminal-callback proposals; rates are pooled fallback/proposal ratios. Per-run and aggregate counts are in `fmqa_proposal_counts.csv` and `fmqa_fallback_by_family.csv`. Uniform fallback supplies most process proposals, so the results characterize the combined learned-proposal and fallback procedure. The added uniform control compares the complete FMQA-SA procedure with sampling under the shared protocol; it does not isolate the individual causal contributions of fitting, annealing, and fallback.

Table S2: Recorded FMQA-SA uniform fallback by benchmark family. Counts include proposals for terminal callbacks.

Family	Runs	Proposals	Fallbacks	Rate
Grid	1,050	51,450	9,522	18.51%
CSTR	60	1,350	370	27.41%
Column	10	130	82	63.08%
Dynamic	60	1,460	1,063	72.81%
Small Batch	10	70	58	82.86%
Process total	140	3,010	1,573	52.26%

S4.3 Terminal-callback solutions and final summaries

Process success is defined by the wrapper's best accepted configuration, including callback B, rather than by the native optimizer's returned incumbent. The non-BAM runners record callback B before raising `BudgetStop`; `worker()` records normal budget completion, and `summarize()` selects the minimum `raw_objective` among `accepted=True` records. Its coordinates and objective are stored as `best_point` and `best_accepted_objective` in `runs.csv`.

The following terminal audit concerns the original 560-run comparison containing RBFOpt (solver): 291 hits, 268 misses, and one execution failure. The new RBFOpt and uniform terminal observations are separately preserved in the additional archive and the manuscript companion table `new_terminal_hits.csv`. Twenty hits first occur at callback B: six values return normally to BAM, while ten RBFOpt (solver) and four FMQA-SA hits are preserved only in the wrapper summary. For these 14 terminal-only cases (Table S3), the accepted callback coordinates and objective match the final summary and reference exactly. All have `budget_stop` and a null `backend_result_at_budget`; their native returned incumbents are therefore unavailable. Hit counts, completion profiles, and restart estimates consistently use the wrapper metric.

Table S3: The 14 terminal-only target successes preserved in the original final wrapper summaries. In every row, the displayed coordinates equal both the terminal configuration ξ_B and the final wrapper configuration ξ_{wrap}; the displayed objective equals the accepted terminal value f_B, the final wrapper value f_{wrap}, and f_{ref}. Native return is unavailable. C and N identify the source archives below.

Case	B	Method	Seed	$\xi_B = \xi_{\text{wrap}}$	$f_B = f_{\text{wrap}} = f_{\text{ref}}$	Source
CSTR 10	15	RBFOpt (solver)	1	$(10, 10)$	2.889536968602369	C
CSTR 10	15	RBFOpt (solver)	5	$(10, 10)$	2.889536968602369	C
CSTR 10	15	RBFOpt (solver)	6	$(10, 10)$	2.889536968602369	C
CSTR 5	8	FMQA-SA	5	$(5, 5)$	3.0620145766081155	C
CSTR 5	8	RBFOpt (solver)	3	$(5, 5)$	3.0620145766081155	C
CSTR 5	8	RBFOpt (solver)	7	$(5, 5)$	3.0620145766081155	C
Column	16	FMQA-SA	8	$(13, 4)$	19346.124801006445	N
Column	16	RBFOpt (solver)	3	$(13, 4)$	19346.124801006445	N
Dynamic	12	FMQA-SA	7	$(1, 4)$	-212.00878715666533	N
Dynamic	12	RBFOpt (solver)	7	$(1, 4)$	-212.00878715666533	N
Dynamic	15	FMQA-SA	3	$(1, 4)$	-212.00878715666533	N
Small Batch	10	RBFOpt (solver)	5	$(2, 2, 1)$	167427.6571147878	N
Small Batch	10	RBFOpt (solver)	7	$(2, 2, 1)$	167427.6571147878	N
Small Batch	10	RBFOpt (solver)	9	$(2, 2, 1)$	167427.6571147878	N

The target tolerance is $10^{-5} \max\{1, |f_{\text{ref}}|\}$; equality in Table S3 is checked before rounding. Sources C and N are `ipopt_dfo_cstr_20261015_174637_r1dzn5tk.zip` and `ipopt_dfo_non_cstr_20261015_172428_o671edu_.zip`, respectively, under `results/archives/`. The companion `terminal_callback_recovery.json` links each case to its original `callbacks.jsonl`, `result.json`, and `runs.csv` row, records the preterminal and terminal objectives, and identifies the executed source copies and archive/source hashes. Repository and experimental revisions are given in Section S8.

S4.4 Stopping diagnostics

The retained historical stopping audit uses 4,200 Grid runs and 560 process runs, with RBFOpt explicitly referring to the solver variant. All new RBFOpt and sampling runs reach their declared callback budgets; their first-hit and post-target counts can be reconstructed from `manuscript/tables/run_records.csv`. Table S4 pools the tested conditions and budgets, with 1,050 runs per Grid row and 140 per process row; `stopping_diagnostics.json` provides run-level and condition-level records.

Budget denotes reaching the callback limit, including BAM's native function-call limit. For Grid RBFOpt and FMQA-SA, this classification uses recorded callback counts because a separate native stopping reason was not stored. Explicit Grid DSDA target/local termination reasons take precedence even when the final callback equals B. No other native early termination or time-limit termination was identified in the selected records.

Grid DSDA checks its target after a local-search run rather than after every callback, so target-based termination can include post-target evaluations. The single process failure is Column DSDA at $B = 16$, seed 0: all three initial configurations were rejected by logic or preprocessing checks, leaving no accepted starting point; no reseeding was performed.

Table S4: Stopping diagnostics from the selected execution records. The final column is the median number of additional callbacks after first target attainment among successful runs, with the contributing number of successful runs in parentheses.

Method	Budget	Local	Target	Failure	Callbacks	After target
Grid DSDA	536	212	302	0	38,033	10 (382)
Grid RBFOpt (solver)	1050	0	0	0	54,600	40 (661)
Grid BAM	1050	0	0	0	54,600	45 (626)
Grid FMQA-SA	1050	0	0	0	54,600	31 (370)
Process DSDA	73	66	0	1	2,532	6 (75)
Process RBFOpt (solver)	140	0	0	0	3,430	14 (127)
Process BAM	140	0	0	0	3,430	7 (47)
Process FMQA-SA	140	0	0	0	3,430	7.5 (42)

Post-target counts exclude the first target callback and are undefined for target misses and execution failures. These diagnostics do not isolate causal runtime contributions, and total runtime was not prorated to estimate first-target time.

S5 Process model scale, units, and discrete decisions

Counts below describe the unfixed original GDP before reformulation or presolve, including all active disjunct branches, not the smaller NLP for a particular selected configuration. Fixed continuous variables remain in the continuous-variable count. Dynamic is discretized with 30 finite elements per stage and three Radau points. Binary counts refer to the indicator variables already constructed by the model, before Boolean-to-binary conversion. Section S9 reconciles these counts with the configuration-specific CSTR15 feasibility check. All nine models have zero additional nonbinary integer variables.

Table S5: Original process-model sizes. n_c: continuous variables; n_b: binary variables; m_a: algebraic constraints; m_l: logical constraints; n_d: disjunctions.

Model	n_c	n_b	m_a	m_l	n_d
CSTR 5	56	20	100	12	10
CSTR 10	106	40	195	22	20
CSTR 15	156	60	290	32	30
CSTR 20	206	80	385	42	40
CSTR 25	256	100	480	52	50
CSTR 30	306	120	575	62	60
Column	446	30	637	16	15
Small Batch	19	18	34	3	9
Dynamic	1647	27	3276	31	9

For CSTR, the cutoff/feed and recycle choices determine reactor/bypass and recycle disjuncts; the objective is total reactor volume in litres. Column selects reflux/boilup positions and the implied tray configuration. Its implemented objective is $10^3(Q_c + Q_b) + 10^3 n_{\text{active}}$, a weighted cost proxy labelled in dollars by the source model; it is not a newly estimated annualized economic cost. Small Batch selects the parallel-unit counts for three stages, fixing the discrete unit-number alternatives; its source objective is investment cost in dollars. No currency-year conversion is applied. Dynamic's two switching-stage choices fix each stage's mode and leave the continuous state trajectories to the discretized NLP. Its objective is $-\int_0^9 x(t)^2\,dt$, so minimization favors a larger integrated squared state. State and time are abstract benchmark quantities without a physical unit calibration in the implementation. The example adds mode sequencing and dynamic differential constraints to the static equipment/configuration cases; it is not a plant-specific economic metric.

Column uses the predeclared historical lookup-study box $\{7,\dots,15\} \times \{1,\dots,9\}$ (81 points). This is a restricted comparison domain within the full 225-point GDP box, not a proof that all outside configurations are infeasible. The box was chosen as an experimental scope restriction without a specific engineering or optimization-based selection criterion. Consequently, the DFO findings are conditional on this restricted domain. All four DFO methods use the same box; the algebraic references use the original GDP domain.

S6 Grid inventory and target-set sizes

All DFO methods receive the integer pair $(i, j) \in \{0,\dots,30\}^2$. The one-based external-variable convention in the main paper corresponds to $(\xi_1, \xi_2) = (i + 1, j + 1)$; this is a relabelling, not an offset applied by the Grid optimizers. Physical coordinates follow Equation (S1). FMQA-SA uses

five unsigned bits per Grid coordinate; decoded coordinate values equal to 31 are rejected as out of bounds. The normalized tolerance 10^{-5} equals $10^{-5}(U - L)$ in raw objective units. Target-set sizes therefore vary across functions even though the normalized tolerance is shared.

The inventory below is calculated from all 961 implemented raw samples. Constants are rounded for display; full precision, physical coordinates, and complete minimum/target lists are in `grid_inventory.json`. $n_{\min}^{\mathrm{fp}}$ counts exact ties at the smallest floating-point raw sample, and n_{target} counts normalized samples at most 10^{-5}. Floating arithmetic may split analytic ties: Rosenbrock has the two exact minimizers stated in the main paper, although only one sample equals the floating minimum bit for bit. For Xin-She Yang 3 (historical variant), underflow gives 12 exact stored minima, and 870 grid points meet the target. Neither exact stored zeros nor target membership imply that the corresponding continuous analytical function is zero there.

Table S6: Enumerated grid normalization constants and target-set sizes.

Function	L	U	$n_{\min}^{\mathrm{fp}}$	n_{target}
Ackley	4.440892e-16	14.24694	1	1
Alpine 1	0	17.42973	1	1
Alpine 2	-896.0502	789.5396	1	1
Bohachevsky 2	0	30000	1	1
Three-hump camel	0	2047.917	1	1
Csendes (squared variant)	0	16.14791	1	36
Deflected corrugated spring	-1	5.698269	1	1
Drop-Wave	-1	-1.314513e-07	1	1
Easom	-0.8953676	0.008461561	1	1
Griewank	0	180.0121	1	1
Hölder table	-18.13802	-0	2	4
Hosaki	-2.345812	0.5413411	1	1
Lévy	0.1140618	95.38281	1	2
Lévy 13	0.3055556	307.8611	1	4
Michalewicz	-1.80107	-0	1	1
Rastrigin	0	77.84942	1	1
Rosenbrock	0.006419753	2509	1	2
Shubert (historical variant)	-11.03782	10.72027	1	1
Styblinski–Tang	-1	3.205128	1	1
Sum of squares	0	300	1	1
Xin-She Yang 3 (historical variant)	0	3	12	870

S6.1 Stored minimizing grid indices

The indices below attain the stored floating-point minima, subject to the tie and underflow qualifications above.

Table S7: Indices attaining the smallest stored raw value.

Function	Minimizing (i, j)
Ackley	(15,15)
Alpine 1	(0,0)
Alpine 2	(27,27)
Bohachevsky 2	(15,15)
Three-hump camel	(15,15)
Csendes (squared variant)	(15,15)
Deflected corrugated spring	(15,15)
Drop-Wave	(15,15)
Easom	(20,20)
Griewank	(15,15)
Hölder table	(3,1), (27,1)
Hosaki	(24,10)
Lévy	(16,17)
Lévy 13	(17,17)
Michalewicz	(21,15)
Rastrigin	(15,15)
Rosenbrock	(22,14)
Shubert (historical variant)	(17,22)
Styblinski–Tang	(6,6)
Sum of squares	(15,15)
Xin-She Yang 3 (historical variant)	(0,0), (0,1), (0,29), (0,30), (1,0), (1,30), (29,0), (29,30), (30,0), (30,1), (30,29), (30,30)

S7 Success-rate uncertainty and restart interpretation

For each problem, method, and budget, let h be the number of target hits among $n = 10$ seeds, retaining execution failures as non-successes. Process hits follow the wrapper metric in Section S4.3; the same indicators enter the TTS99 calculation.

We report two-sided 95% Clopper–Pearson intervals for the binomial success probability (Clopper and Pearson, 1934):

$$p_L = \begin{cases} 0, & h = 0, \\ B^{-1}_{0.025}(h, n-h+1), & h > 0, \end{cases} \qquad p_U = \begin{cases} 1, & h = n, \\ B^{-1}_{0.975}(h+1, n-h), & h < n, \end{cases}$$

where B^{-1} denotes the beta quantile. Coverage is conditional on treating the seed-level outcomes as independent Bernoulli trials under the fixed protocol. The intervals are pointwise, not simultaneous across comparisons.

Ten observed successes give $[0.6915, 1]$; zero give $[0, 0.3085]$. Thus neither empirical endpoint establishes a population probability of one or zero. For ten hits, the plug-in restart count is one, whereas substituting the lower bound 0.6915 yields four runs. This illustrates uncertainty propagation and does not create a combined confidence guarantee. TTS99 multiplies the median full-run cost by a plug-in independent-restart count, equal to one for $h = 10$ and infinity for $h = 0$. It is a descriptive summary under independence and representative-cost assumptions, not a runtime percentile or a 99% guarantee; no first-hit time is inferred from full-run overhead.

The following tables cover 70 primary process groups and 630 primary Grid problem–method–budget groups. Comparisons with RBFOpt (solver) are reported in Section S10. Cells report $h\,[p_L, p_U]$, with limits rounded to two decimals; displayed endpoints may be rounded interior bounds. The primary counts and interval tables are regenerated from `manuscript/tables/run_records.csv`; the original companion `success_intervals.csv` is retained as a historical solver-variant artifact. Seed indices remain paired across methods within each problem/budget, but intervals are calculated separately for each group, not by pooling the heterogeneous 21-function suite.

Table S8: Process target hits and pointwise 95% success-probability intervals. All failures remain in the ten-run denominator.

Case	B	DSDA	RBFOpt	BAM	FMQA-SA	Uniform
Column	16	5 [0.19, 0.81]	5 [0.19, 0.81]	0 [0.00, 0.31]	4 [0.12, 0.74]	0 [0.00, 0.31]
CSTR 5	8	2 [0.03, 0.56]	10 [0.69, 1.00]	10 [0.69, 1.00]	3 [0.07, 0.65]	4 [0.12, 0.74]
CSTR 10	15	2 [0.03, 0.56]	9 [0.55, 1.00]	1 [0.00, 0.45]	2 [0.03, 0.56]	1 [0.00, 0.45]
CSTR 15	22	2 [0.03, 0.56]	8 [0.44, 0.97]	1 [0.00, 0.45]	1 [0.00, 0.45]	2 [0.03, 0.56]
CSTR 20	29	3 [0.07, 0.65]	9 [0.55, 1.00]	2 [0.03, 0.56]	1 [0.00, 0.45]	2 [0.03, 0.56]
CSTR 25	36	3 [0.07, 0.65]	10 [0.69, 1.00]	2 [0.03, 0.56]	1 [0.00, 0.45]	1 [0.00, 0.45]
CSTR 30	43	3 [0.07, 0.65]	10 [0.69, 1.00]	2 [0.03, 0.56]	1 [0.00, 0.45]	1 [0.00, 0.45]
Small Batch	10	8 [0.44, 0.97]	5 [0.19, 0.81]	2 [0.03, 0.56]	6 [0.26, 0.88]	4 [0.12, 0.74]
Dynamic	12	4 [0.12, 0.74]	10 [0.69, 1.00]	0 [0.00, 0.31]	2 [0.03, 0.56]	0 [0.00, 0.31]
Dynamic	15	5 [0.19, 0.81]	10 [0.69, 1.00]	3 [0.07, 0.65]	3 [0.07, 0.65]	3 [0.07, 0.65]
Dynamic	20	8 [0.44, 0.97]	10 [0.69, 1.00]	3 [0.07, 0.65]	3 [0.07, 0.65]	3 [0.07, 0.65]
Dynamic	30	10 [0.69, 1.00]	10 [0.69, 1.00]	4 [0.12, 0.74]	4 [0.12, 0.74]	4 [0.12, 0.74]
Dynamic	37	10 [0.69, 1.00]	10 [0.69, 1.00]	7 [0.35, 0.93]	5 [0.19, 0.81]	5 [0.19, 0.81]
Dynamic	50	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	6 [0.26, 0.88]	6 [0.26, 0.88]

Table S9: Grid target hits and pointwise 95% success-probability intervals at $B = 10$. Each cell is $h\,[p_L, p_U]$ from ten seeds.

Function	DSDA	RBFOpt	BAM	FMQA-SA	Uniform	BAM-prefix control
Ackley	0 [0.00, 0.31]	4 [0.12, 0.74]	10 [0.69, 1.00]	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]
Alpine1	0 [0.00, 0.31]	10 [0.69, 1.00]	10 [0.69, 1.00]	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]
Alpine2	1 [0.00, 0.45]	1 [0.00, 0.45]	1 [0.00, 0.45]	1 [0.00, 0.45]	1 [0.00, 0.45]	1 [0.00, 0.45]
Bohachevsky2	0 [0.00, 0.31]	2 [0.03, 0.56]	10 [0.69, 1.00]	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]
Camel3	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]
Csendes	0 [0.00, 0.31]	2 [0.03, 0.56]	10 [0.69, 1.00]	1 [0.00, 0.45]	2 [0.03, 0.56]	10 [0.69, 1.00]
Deflected corrugated spring	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]
Dropwave	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]
Easom	0 [0.00, 0.31]	1 [0.00, 0.45]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]
Griewank	0 [0.00, 0.31]	2 [0.03, 0.56]	10 [0.69, 1.00]	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]
Holder Table	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	1 [0.00, 0.45]	0 [0.00, 0.31]	0 [0.00, 0.31]
Hosaki	1 [0.00, 0.45]	3 [0.07, 0.65]	0 [0.00, 0.31]	1 [0.00, 0.45]	0 [0.00, 0.31]	0 [0.00, 0.31]
Levy	0 [0.00, 0.31]	1 [0.00, 0.45]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]
Levy13	0 [0.00, 0.31]	2 [0.03, 0.56]	0 [0.00, 0.31]	2 [0.03, 0.56]	1 [0.00, 0.45]	0 [0.00, 0.31]
Michalewicz	1 [0.00, 0.45]	1 [0.00, 0.45]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]
Rastrigin	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]
Rosenbrock	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]
Shubert	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]
Styblinski Tang	0 [0.00, 0.31]	1 [0.00, 0.45]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]
Sum Of Squares	0 [0.00, 0.31]	2 [0.03, 0.56]	10 [0.69, 1.00]	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]
Xin She Yang3	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]

Table S10: Grid target hits and pointwise 95% success-probability intervals at $B = 30$. Each cell is $h\,[p_L, p_U]$ from ten seeds.

Function	DSDA	RBFOpt	BAM	FMQA-SA	Uniform	BAM-prefix control
Ackley	0 [0.00, 0.31]	10 [0.69, 1.00]	10 [0.69, 1.00]	1 [0.00, 0.45]	0 [0.00, 0.31]	10 [0.69, 1.00]
Alpine1	1 [0.00, 0.45]	10 [0.69, 1.00]	10 [0.69, 1.00]	3 [0.07, 0.65]	0 [0.00, 0.31]	10 [0.69, 1.00]
Alpine2	1 [0.00, 0.45]	1 [0.00, 0.45]	1 [0.00, 0.45]	1 [0.00, 0.45]	1 [0.00, 0.45]	2 [0.03, 0.56]
Bohachevsky2	6 [0.26, 0.88]	10 [0.69, 1.00]	10 [0.69, 1.00]	0 [0.00, 0.31]	1 [0.00, 0.45]	10 [0.69, 1.00]
Camel3	1 [0.00, 0.45]	9 [0.55, 1.00]	10 [0.69, 1.00]	0 [0.00, 0.31]	2 [0.03, 0.56]	10 [0.69, 1.00]
Csendes	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	8 [0.44, 0.97]	6 [0.26, 0.88]	10 [0.69, 1.00]
Deflected corrugated spring	0 [0.00, 0.31]	1 [0.00, 0.45]	10 [0.69, 1.00]	2 [0.03, 0.56]	0 [0.00, 0.31]	10 [0.69, 1.00]
Dropwave	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]
Easom	1 [0.00, 0.45]	3 [0.07, 0.65]	0 [0.00, 0.31]	1 [0.00, 0.45]	0 [0.00, 0.31]	0 [0.00, 0.31]
Griewank	9 [0.55, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	2 [0.03, 0.56]	0 [0.00, 0.31]	10 [0.69, 1.00]
Holder Table	1 [0.00, 0.45]	1 [0.00, 0.45]	0 [0.00, 0.31]	5 [0.19, 0.81]	1 [0.00, 0.45]	0 [0.00, 0.31]
Hosaki	6 [0.26, 0.88]	10 [0.69, 1.00]	0 [0.00, 0.31]	2 [0.03, 0.56]	0 [0.00, 0.31]	1 [0.00, 0.45]
Levy	2 [0.03, 0.56]	8 [0.44, 0.97]	0 [0.00, 0.31]	2 [0.03, 0.56]	0 [0.00, 0.31]	0 [0.00, 0.31]
Levy13	8 [0.44, 0.97]	10 [0.69, 1.00]	3 [0.07, 0.65]	6 [0.26, 0.88]	3 [0.07, 0.65]	1 [0.00, 0.45]
Michalewicz	5 [0.19, 0.81]	6 [0.26, 0.88]	1 [0.00, 0.45]	1 [0.00, 0.45]	0 [0.00, 0.31]	1 [0.00, 0.45]
Rastrigin	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]
Rosenbrock	1 [0.00, 0.45]	0 [0.00, 0.31]	0 [0.00, 0.31]	2 [0.03, 0.56]	1 [0.00, 0.45]	0 [0.00, 0.31]
Shubert	1 [0.00, 0.45]	1 [0.00, 0.45]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]
Styblinski Tang	6 [0.26, 0.88]	10 [0.69, 1.00]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]
Sum Of Squares	6 [0.26, 0.88]	10 [0.69, 1.00]	10 [0.69, 1.00]	1 [0.00, 0.45]	0 [0.00, 0.31]	10 [0.69, 1.00]
Xin She Yang3	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]

Table S11: Grid target hits and pointwise 95% success-probability intervals at $B = 50$. Each cell is $h\,[p_L, p_U]$ from ten seeds.

Function	DSDA	RBFOpt	BAM	FMQA-SA	Uniform	BAM-prefix control
Ackley	0 [0.00, 0.31]	10 [0.69, 1.00]	10 [0.69, 1.00]	4 [0.12, 0.74]	0 [0.00, 0.31]	10 [0.69, 1.00]
Alpine1	1 [0.00, 0.45]	10 [0.69, 1.00]	10 [0.69, 1.00]	3 [0.07, 0.65]	0 [0.00, 0.31]	10 [0.69, 1.00]
Alpine2	1 [0.00, 0.45]	1 [0.00, 0.45]	1 [0.00, 0.45]	2 [0.03, 0.56]	1 [0.00, 0.45]	2 [0.03, 0.56]
Bohachevsky2	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	2 [0.03, 0.56]	1 [0.00, 0.45]	10 [0.69, 1.00]
Camel3	1 [0.00, 0.45]	10 [0.69, 1.00]	10 [0.69, 1.00]	1 [0.00, 0.45]	2 [0.03, 0.56]	10 [0.69, 1.00]
Csendes	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	9 [0.55, 1.00]	10 [0.69, 1.00]
Deflected corrugated spring	0 [0.00, 0.31]	1 [0.00, 0.45]	10 [0.69, 1.00]	2 [0.03, 0.56]	0 [0.00, 0.31]	10 [0.69, 1.00]
Dropwave	0 [0.00, 0.31]	2 [0.03, 0.56]	10 [0.69, 1.00]	1 [0.00, 0.45]	0 [0.00, 0.31]	10 [0.69, 1.00]
Easom	2 [0.03, 0.56]	5 [0.19, 0.81]	0 [0.00, 0.31]	3 [0.07, 0.65]	0 [0.00, 0.31]	1 [0.00, 0.45]
Griewank	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	3 [0.07, 0.65]	0 [0.00, 0.31]	10 [0.69, 1.00]
Holder Table	1 [0.00, 0.45]	10 [0.69, 1.00]	1 [0.00, 0.45]	8 [0.44, 0.97]	2 [0.03, 0.56]	0 [0.00, 0.31]
Hosaki	7 [0.35, 0.93]	10 [0.69, 1.00]	2 [0.03, 0.56]	3 [0.07, 0.65]	0 [0.00, 0.31]	2 [0.03, 0.56]
Levy	2 [0.03, 0.56]	10 [0.69, 1.00]	1 [0.00, 0.45]	3 [0.07, 0.65]	0 [0.00, 0.31]	1 [0.00, 0.45]
Levy13	8 [0.44, 0.97]	10 [0.69, 1.00]	5 [0.19, 0.81]	7 [0.35, 0.93]	3 [0.07, 0.65]	1 [0.00, 0.45]
Michalewicz	7 [0.35, 0.93]	9 [0.55, 1.00]	1 [0.00, 0.45]	4 [0.12, 0.74]	0 [0.00, 0.31]	2 [0.03, 0.56]
Rastrigin	0 [0.00, 0.31]	2 [0.03, 0.56]	10 [0.69, 1.00]	0 [0.00, 0.31]	0 [0.00, 0.31]	10 [0.69, 1.00]
Rosenbrock	1 [0.00, 0.45]	2 [0.03, 0.56]	0 [0.00, 0.31]	2 [0.03, 0.56]	1 [0.00, 0.45]	0 [0.00, 0.31]
Shubert	1 [0.00, 0.45]	2 [0.03, 0.56]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]
Styblinski Tang	7 [0.35, 0.93]	10 [0.69, 1.00]	0 [0.00, 0.31]	5 [0.19, 0.81]	0 [0.00, 0.31]	0 [0.00, 0.31]
Sum Of Squares	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	3 [0.07, 0.65]	0 [0.00, 0.31]	10 [0.69, 1.00]
Xin She Yang3	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]

Table S12: Grid target hits and pointwise 95% success-probability intervals at $B = 70$. Each cell is $h\,[p_L, p_U]$ from ten seeds.

Function	DSDA	RBFOpt	BAM	FMQA-SA	Uniform	BAM-prefix control
Ackley	0 [0.00, 0.31]	10 [0.69, 1.00]	10 [0.69, 1.00]	5 [0.19, 0.81]	1 [0.00, 0.45]	10 [0.69, 1.00]
Alpine1	1 [0.00, 0.45]	10 [0.69, 1.00]	10 [0.69, 1.00]	4 [0.12, 0.74]	1 [0.00, 0.45]	10 [0.69, 1.00]
Alpine2	1 [0.00, 0.45]	2 [0.03, 0.56]	1 [0.00, 0.45]	2 [0.03, 0.56]	1 [0.00, 0.45]	2 [0.03, 0.56]
Bohachevsky2	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	5 [0.19, 0.81]	1 [0.00, 0.45]	10 [0.69, 1.00]
Camel3	8 [0.44, 0.97]	10 [0.69, 1.00]	10 [0.69, 1.00]	2 [0.03, 0.56]	2 [0.03, 0.56]	10 [0.69, 1.00]
Csendes	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	9 [0.55, 1.00]	10 [0.69, 1.00]
Deflected corrugated spring	0 [0.00, 0.31]	1 [0.00, 0.45]	10 [0.69, 1.00]	3 [0.07, 0.65]	1 [0.00, 0.45]	10 [0.69, 1.00]
Dropwave	0 [0.00, 0.31]	2 [0.03, 0.56]	10 [0.69, 1.00]	2 [0.03, 0.56]	0 [0.00, 0.31]	10 [0.69, 1.00]
Easom	3 [0.07, 0.65]	7 [0.35, 0.93]	0 [0.00, 0.31]	3 [0.07, 0.65]	0 [0.00, 0.31]	1 [0.00, 0.45]
Griewank	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	5 [0.19, 0.81]	1 [0.00, 0.45]	10 [0.69, 1.00]
Holder Table	1 [0.00, 0.45]	10 [0.69, 1.00]	2 [0.03, 0.56]	8 [0.44, 0.97]	3 [0.07, 0.65]	0 [0.00, 0.31]
Hosaki	9 [0.55, 1.00]	10 [0.69, 1.00]	3 [0.07, 0.65]	5 [0.19, 0.81]	0 [0.00, 0.31]	3 [0.07, 0.65]
Levy	2 [0.03, 0.56]	10 [0.69, 1.00]	1 [0.00, 0.45]	4 [0.12, 0.74]	0 [0.00, 0.31]	1 [0.00, 0.45]
Levy13	8 [0.44, 0.97]	10 [0.69, 1.00]	5 [0.19, 0.81]	8 [0.44, 0.97]	5 [0.19, 0.81]	1 [0.00, 0.45]
Michalewicz	7 [0.35, 0.93]	9 [0.55, 1.00]	1 [0.00, 0.45]	7 [0.35, 0.93]	0 [0.00, 0.31]	2 [0.03, 0.56]
Rastrigin	0 [0.00, 0.31]	3 [0.07, 0.65]	10 [0.69, 1.00]	1 [0.00, 0.45]	0 [0.00, 0.31]	10 [0.69, 1.00]
Rosenbrock	2 [0.03, 0.56]	5 [0.19, 0.81]	0 [0.00, 0.31]	5 [0.19, 0.81]	1 [0.00, 0.45]	0 [0.00, 0.31]
Shubert	1 [0.00, 0.45]	2 [0.03, 0.56]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]	0 [0.00, 0.31]
Styblinski Tang	8 [0.44, 0.97]	10 [0.69, 1.00]	9 [0.55, 1.00]	7 [0.35, 0.93]	0 [0.00, 0.31]	0 [0.00, 0.31]
Sum Of Squares	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	6 [0.26, 0.88]	0 [0.00, 0.31]	10 [0.69, 1.00]
Xin She Yang3	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]

Table S13: Grid target hits and pointwise 95% success-probability intervals at $B = 100$. Each cell is $h\,[p_L, p_U]$ from ten seeds.

Function	DSDA	RBFOpt	BAM	FMQA-SA	Uniform	BAM-prefix control
Ackley	0 [0.00, 0.31]	10 [0.69, 1.00]	10 [0.69, 1.00]	8 [0.44, 0.97]	1 [0.00, 0.45]	10 [0.69, 1.00]
Alpine1	1 [0.00, 0.45]	10 [0.69, 1.00]	10 [0.69, 1.00]	5 [0.19, 0.81]	1 [0.00, 0.45]	10 [0.69, 1.00]
Alpine2	1 [0.00, 0.45]	3 [0.07, 0.65]	1 [0.00, 0.45]	2 [0.03, 0.56]	1 [0.00, 0.45]	2 [0.03, 0.56]
Bohachevsky2	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	9 [0.55, 1.00]	1 [0.00, 0.45]	10 [0.69, 1.00]
Camel3	9 [0.55, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	3 [0.07, 0.65]	2 [0.03, 0.56]	10 [0.69, 1.00]
Csendes	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]
Deflected corrugated spring	0 [0.00, 0.31]	1 [0.00, 0.45]	10 [0.69, 1.00]	3 [0.07, 0.65]	3 [0.07, 0.65]	10 [0.69, 1.00]
Dropwave	0 [0.00, 0.31]	2 [0.03, 0.56]	10 [0.69, 1.00]	4 [0.12, 0.74]	0 [0.00, 0.31]	10 [0.69, 1.00]
Easom	3 [0.07, 0.65]	7 [0.35, 0.93]	0 [0.00, 0.31]	3 [0.07, 0.65]	0 [0.00, 0.31]	1 [0.00, 0.45]
Griewank	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	2 [0.03, 0.56]	10 [0.69, 1.00]
Holder Table	1 [0.00, 0.45]	10 [0.69, 1.00]	6 [0.26, 0.88]	8 [0.44, 0.97]	4 [0.12, 0.74]	1 [0.00, 0.45]
Hosaki	10 [0.69, 1.00]	10 [0.69, 1.00]	3 [0.07, 0.65]	6 [0.26, 0.88]	0 [0.00, 0.31]	3 [0.07, 0.65]
Levy	2 [0.03, 0.56]	10 [0.69, 1.00]	5 [0.19, 0.81]	4 [0.12, 0.74]	0 [0.00, 0.31]	2 [0.03, 0.56]
Levy13	8 [0.44, 0.97]	10 [0.69, 1.00]	9 [0.55, 1.00]	9 [0.55, 1.00]	6 [0.26, 0.88]	2 [0.03, 0.56]
Michalewicz	7 [0.35, 0.93]	10 [0.69, 1.00]	2 [0.03, 0.56]	7 [0.35, 0.93]	0 [0.00, 0.31]	2 [0.03, 0.56]
Rastrigin	0 [0.00, 0.31]	4 [0.12, 0.74]	10 [0.69, 1.00]	5 [0.19, 0.81]	1 [0.00, 0.45]	10 [0.69, 1.00]
Rosenbrock	3 [0.07, 0.65]	10 [0.69, 1.00]	0 [0.00, 0.31]	6 [0.26, 0.88]	2 [0.03, 0.56]	1 [0.00, 0.45]
Shubert	1 [0.00, 0.45]	4 [0.12, 0.74]	1 [0.00, 0.45]	1 [0.00, 0.45]	1 [0.00, 0.45]	0 [0.00, 0.31]
Styblinski Tang	8 [0.44, 0.97]	10 [0.69, 1.00]	10 [0.69, 1.00]	7 [0.35, 0.93]	0 [0.00, 0.31]	0 [0.00, 0.31]
Sum Of Squares	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	9 [0.55, 1.00]	0 [0.00, 0.31]	10 [0.69, 1.00]
Xin She Yang3	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]

S8 Data sources and reproducibility

The manuscript and supplement cite the snapshot of `main` at commit `cb70771b2037301bb6d0e38a2362d49b53fcda7d`, accessed October 19, 2026 (SECQUOIA, 2026b). The selected Grid and process analyses are `results/grid_gdp_dfo_20261011` and `results/process_ipopt_bam_v2_20261015` in that snapshot. The process return archives retain their experimental revision `ade67082ad2b09583a8f5594b827c97417e0ea99`; per-job settings and source hashes are supplied in the companion files described in Section S4.

The Process integration manifest links the original archives, runner packages, frozen inputs, revisions, and hashes. The Process reproduction guide and Grid reproduction guide provide verification and table-regeneration commands. The Grid algebraic-run index records all 105 selected problem–method outcomes, including times for executed target misses, statuses, termination labels, and original-result paths and hashes. The companion `figure_sources/` directory provides code, input identifiers, and rendering instructions for manuscript Figures 2, 6, and 7.

S9 Feasibility and bound consistency of the algebraic results

Tables S14 and S15 provide the evidence for the † annotations in manuscript Table 8: all 45 selected runs and nine saved feasible comparison solutions, using the snapshot in Section S8. The companion `algebraic_audit/` directory contains `table8_bound_audit.csv` and `feasible_witnesses.csv`, joined by witness ID, together with 63 byte-preserved original result objects and saved vectors, original-GDP rechecks, and archive provenance. The 63 checked vectors comprise 45 selected-run vectors and 18 earlier RIC/GLOA vectors (nine per method, September 25 archive); the nine displayed witnesses are a subset: seven selected-run vectors and two earlier RIC vectors (Column and Dynamic). Both CSVs link to the raw records by archive-member path and SHA-256 hash.

For each minimization model, let f_w be the recorded objective of its best independently checked saved feasible solution, and let L denote a reported lower bound. We apply the same rule to all five methods:

$$\tau_w = 10^{-5}\max\{1, |f_w|\}, \qquad \Delta = L - f_w, \qquad \rho = \Delta/\tau_w, \qquad \dagger \iff \rho > 1. \tag{S24}$$

Missing lower bounds remain missing, not zero. The CSV stores unrounded reported and reevaluated objectives, L, original-model violation, f_w, τ_w, Δ, and ρ. Reported values retain their original decimal strings; calculated values use binary64 round-trip representations. Only the violations and ratios displayed in this document are rounded. For CSTR20, the recorded and reevaluated witness objectives are 2.7994063086813563 and 2.7994063086813568, respectively; both give the same bound classification.

Saved vectors are loaded into the original GDP to reevaluate objectives and check active constraints, variable bounds, integrality, disjunctions, and logical validity without optimization. Violations are unscaled original-equation residuals, not reformulation residuals. All 63 vectors have defined expressions and satisfy the 10^{-5} threshold; the largest violation is $8.947372728933256 \times 10^{-7}$. This numerical check establishes neither exact feasibility nor global optimality. Historical witnesses do not replace selected-run objectives, times, or termination labels.

The uniform rule identifies 11 bound conflicts: the six CSTR GLOA runs, Column Big-M, and CSTR5, CSTR10, CSTR15, and CSTR30 Hull. These conflicting bounds and `optimal` reports provide no independent global certificate or intrinsic method ranking; their cause in GDPopt or its subsolvers remains unresolved.

Table S14: Run-level evidence for manuscript Table 8. Objectives and lower bounds retain their stored digits; $v_{\max}$ and ρ are rounded for display. A dash means no lower bound was reported; a dagger is assigned from the unrounded $\rho > 1$ test. W denotes the witness ID in Table S15. These eleven flags are operational numerical-consistency categories, not eleven equally strong demonstrations of invalid mathematical bounds. The three smaller Hull discrepancies (CSTR5, CSTR10, and CSTR15) are particularly sensitive to numerical feasibility and tolerance conventions.

ID	Model	Method	Reported objective	Reported L	$v_{\max}$	W	ρ
A01	CSTR5	RIC	3.06198395705692	3.06198395705692	6.645e-07	W01	0
A02	CSTR5	GLOA	3.1301981330832502	3.1301981330832502	1.425e-11	W01	2227.78†
A03	CSTR5	LOA	3.06198395705692	3.06198395705692	6.645e-07	W01	0
A04	CSTR5	Big-M	3.0620135966919806	3.062014538351392	9.699e-08	W01	0.998741
A05	CSTR5	Hull	3.06201457880245	3.06201457880245	6.836e-10	W01	1.00006†
A06	CSTR10	RIC	2.8895080745062898	2.8895080745062898	9.243e-08	W02	0.852127
A07	CSTR10	GLOA	3.44796403174898	3.44796403174898	2.169e-11	W02	19328†
A08	CSTR10	LOA	2.8895080745062898	2.8895080745062898	9.243e-08	W02	0.852127
A09	CSTR10	Big-M	2.8894834524326347	2.889483452432633	3.941e-07	W02	-6.14766e-11
A10	CSTR10	Hull	2.889535836078951	2.889535836078951	5.265e-07	W02	1.81291†
A11	CSTR15	RIC	2.9660174126412966	—	1.300e-07	W03	—
A12	CSTR15	GLOA	3.522552007803194	3.522552007803194	1.306e-07	W03	24500.4†
A13	CSTR15	LOA	2.82935061823614	—	5.634e-07	W03	—
A14	CSTR15	Big-M	2.8293940484587314	0.094	1.064e-09	W03	-96677.7
A15	CSTR15	Hull	2.8293940778489666	2.829394077848967	1.101e-09	W03	1.53603†
A16	CSTR20	RIC	2.9095041668290293	—	1.078e-07	W04	—
A17	CSTR20	GLOA	3.6161229932346584	3.6161229932346584	8.316e-12	W04	29174.6†
A18	CSTR20	LOA	2.88950807452168	—	9.933e-08	W04	—
A19	CSTR20	Big-M	2.7994063086813563	0.094	8.937e-10	W04	-96642.1
A20	CSTR20	Hull	2.799406444190852	1.331416004874146	2.945e-09	W04	-52439.3
A21	CSTR25	RIC	2.859437014149684	—	8.587e-08	W05	—
A22	CSTR25	GLOA	3.647818131769651	3.647818131769651	8.947e-07	W05	31146.5†
A23	CSTR25	LOA	2.7951054674535083	—	5.330e-08	W05	—
A24	CSTR25	Big-M	2.781482623276432	0.644872879543012	1.322e-09	W05	-76815.5
A25	CSTR25	Hull	2.781482618820502	0.903635565082982	6.445e-10	W05	-67512.4
A26	CSTR30	RIC	2.837952314572675	—	7.561e-08	W06	—
A27	CSTR30	GLOA	3.154308073141742	3.154308073141742	4.528e-08	W06	13891.6†
A28	CSTR30	LOA	2.7844377336170396	—	4.731e-08	W06	—
A29	CSTR30	Big-M	2.7695697953461833	0.094	9.948e-09	W06	-96606
A30	CSTR30	Hull	2.8093862310789772	2.809386231078974	8.796e-07	W06	1437.64†

Table S14 continued

ID	Model	Method	Reported objective	Reported L	$v_{\max}$	W	ρ
A31	Column	RIC	22271.54871907239	—	2.461e-09	W07	—
A32	Column	GLOA	22271.54871907239	—	2.461e-09	W07	—
A33	Column	LOA	22271.54871907239	—	2.461e-09	W07	—
A34	Column	Big-M	19853.722763157217	19853.722966464	1.009e-07	W07	2623.77†
A35	Column	Hull	20837.08048191484	15701.345418341163	6.280e-08	W07	-18839.8
A36	Small Batch	RIC	167427.6570040533	167427.65700405455	6.000e-07	W08	1.15395e-05
A37	Small Batch	GLOA	167427.6570040533	167427.65700405455	6.000e-07	W08	1.15395e-05
A38	Small Batch	LOA	167427.6570040533	167427.65700405455	6.000e-07	W08	1.15395e-05
A39	Small Batch	Big-M	167427.6570064299	167427.6570129865	5.966e-07	W08	1.68743e-05
A40	Small Batch	Hull	167427.6569847343	167427.65701108496	5.902e-07	W08	1.57385e-05
A41	Dynamic	RIC	-119.38309935258835	—	3.998e-10	W09	—
A42	Dynamic	GLOA	-119.38309935258835	—	3.998e-10	W09	—
A43	Dynamic	LOA	-119.38309935258835	—	3.998e-10	W09	—
A44	Dynamic	Big-M	-207.62611609437877	-677.0696794915851	1.605e-08	W09	-219359
A45	Dynamic	Hull	-28.596739198040982	-557.0326287276312	1.344e-09	W09	-162740

Table S15: Saved feasible comparison solutions. The CSV links to each full vector and includes original archive/member paths and hashes, reevaluated objective, and unrounded violation.

ID	Model	Method	Recorded f_w	$v_{\max}$	Source
W01	CSTR5	RIC	3.06198395705692	6.645e-07	R2
W02	CSTR10	Big-M	2.8894834524326347	3.941e-07	R1
W03	CSTR15	LOA	2.82935061823614	5.634e-07	R3
W04	CSTR20	Big-M	2.7994063086813563	8.937e-10	R1
W05	CSTR25	Hull	2.781482618820502	6.445e-10	R1
W06	CSTR30	Big-M	2.7695697953461833	9.948e-09	R1
W07	Column	RIC	19346.124801099686	8.527e-14	R1
W08	Small Batch	Hull	167427.6569847343	5.902e-07	R1
W09	Dynamic	RIC	-212.00878715666482	8.313e-13	R1

R1: `results/gdp_comparison_20260925/raw`.
R2: `results/gdp_comparison_20261001/raw`.
R3: `results/gdp_comparison_loa_20261001/raw`.

Within `algebraic_audit/`, `manifest.json` identifies the source archives; `repository_comparison.csv` contains the selected-run results; and `solution_rechecks.json` and `repository_validation.json` provide the original-model checks and validation metadata.

Additional tightened-feasibility check for CSTR15. The original W03 objective and residual in Table S15 were independently reproduced before the follow-up. A separate local IPOPT 3.12.13 solve retained W03's discrete configuration and continuous starting values and targeted an independently checked original-GDP residual of 10^{-8} (SECQUOIA, 2026c). Before that solve, three fixed-Boolean equalities were verified to be exact identities $1 = 1$ and omitted only from a computational clone, reducing its equality count from 158 to 155 while retaining 156 free continuous variables. This preprocessing preserves the feasible set for the fixed configuration; the original GDP, objective, variable bounds, and all other constraints were unchanged. The single invocation terminated normally after 28 iterations and returned objective 2.8293940632857546. Independent reevaluation in a freshly restored original GDP gave maximum unscaled violation $3.3306690738754696 \times 10^{-16}$, passing the 10^{-8} target. The check covered 306 applicable algebraic constraints, variable bounds, integrality, disjunctions, and 32 original logical conditions. Table S5 counts 290 algebraic constraints before reformulation; excluding 135 nonselected-branch rows and adding 151 rows encoding logic gives $290 - 135 + 151 = 306$. The 75 checked discrete values comprise Table S5's 60 disjunct-indicator binaries plus 15 feed-selection binaries, all matching W03. The three omitted identities were rechecked with zero residual.

The recorded A12 GLOA lower bound 3.522552007803194 exceeds the tightened objective by 0.6931579445174396, approximately 24,498 times the unchanged comparison tolerance $10^{-5}\max\{1, |f|\} = 2.829394063285755 \times 10^{-5}$. For Hull, the recorded lower bound 2.829394077848967 exceeds this objective by only 1.45632×10^{-8}, giving $\rho \approx 5.14711 \times 10^{-4} < 1$ and no flag for this comparison, versus $\rho \approx 1.536$ with historical witness W03. This result does not establish exact mathematical feasibility, global optimality, the cause of the reported bound, or a corresponding conclusion for other cases.

This additional validation preserves W03 and all historical audit classifications, performance values, and timings. The two earlier option-rejected invocations and the subsequent degrees-of-freedom failure returned no point and remain archived. The new vector, identity proofs, solver log, original-GDP residuals, and source hashes are archived separately at the cited revision (SECQUOIA, 2026c).

S10 Additional candidate-search and sampling comparisons

The primary comparison uses RBFOpt, uniform sampling, and the Grid BAM-prefix control described in the main text. The 3,430 additional runs comprise 1,190 RBFOpt runs, 1,190 uniform runs, and 1,050 Grid prefix-control runs. All completed their budgets, with 1,558 target hits and 1,872 misses; no failed, interrupted or missing additional slot was replaced. The retained 4,760 historical records remain unchanged, giving 8,190 records in the combined archive. The additional execution-provider revision is `38d917ef75ef6767c2a9f238d88be19bf0b89160`; the later publication/analysis snapshot is identified separately (SECQUOIA, 2026a). The archive contains settings, runtime records, first-hit observations and every charged callback, including unaccepted evaluations and terminal observations. No original NLP was rerun, and each new trajectory uses its own recorded search time and sum of saved point costs plus one-time Dynamic preparation.

S10.1 Comparison with RBFOpt (solver)

RBFOpt (solver) denotes the retained setting with `global_search_method=solver`, while the primary RBFOpt setting uses `global_search_method=genetic`. Nannicini (2021) reported difficulty and higher expense for mathematical-solver candidate search and excluded the MSRSM–solver combination from the principal comparison. This motivates the matched comparison rather than assuming that one option must perform better on the present problems. Only `global_search_method` differs in all 1,190 matched settings comparisons. The retained RBFOpt (solver) records comprise 1,050 Grid runs and 140 Process runs, all with reconstructed completion costs. The original Grid RBFOpt (solver) trace watchdog was disabled. Across all Grid budgets, RBFOpt (solver) and RBFOpt reach 661/1,050 and 658/1,050 targets, respectively; across Process they reach 127/140 and 126/140. These totals summarize heterogeneous conditions and are not independent replications of one common success probability. Relative to RBFOpt (solver), RBFOpt changes Process hit counts from 10 to 9 for CSTR10, 10 to 8 for CSTR15, 4 to 5 for Column, and 9 to 10 for Dynamic B12; the other condition totals agree. The median matched ratio of RBFOpt (solver) to RBFOpt search time is 14.81 for Grid and 6.58 for Process; median reconstructed total-cost ratios are 2.12 and 1.18. These are medians of paired ratios, not ratios of marginal medians. At Grid B=100, RBFOpt (solver) has a median completion estimate of 365.74 s and reaches 172/210 targets, compared with 70.81 s and 171/210 for RBFOpt. RBFOpt substantially reduces the observed search overhead while retaining similar aggregate target counts; these observations do not establish equivalence. The same host, package versions and solver binaries were used, but CPU affinity and the kernel differ, so this is not a simultaneous controlled timing experiment. Other surrogate mathematical-solver calls remain enabled for RBFOpt; the 696 Process Bonmin starts comprise 457 `ok/optimal` returns, 85 `warning/infeasible` returns, and 154 errors. Each error records Pyomo's rejection of a `SolverResults` object with status `error`; the retained logs do not identify the underlying Bonmin cause. RBFOpt 4.3.0 handles an unavailable surrogate minimizer by using its best interpolation node as a reference for an adjusted genetic search; absent or near-duplicate candidates are discarded, and the search advances or restarts under its normal rules. All 30 affected CSTR runs completed their full callback budgets, with 28 reaching the target and two not reaching it. These surrogate events are distinct from process-NLP failures, and their elapsed time remains included in recorded search overhead.

Table S16: RBFOpt (solver) and RBFOpt on Grid. Costs are medians of reconstructed search-completion costs over all 210 runs, including misses.

	RBFOpt (solver)		RBFOpt	
B	Target hits	Cost (s)	Target hits	Cost (s)
10	45/210	5.02	42/210	3.88
30	133/210	24.62	130/210	12.79
50	151/210	75.56	154/210	28.09
70	160/210	159.00	161/210	48.91
100	172/210	365.74	171/210	70.81

Table S17: Matched RBFOpt (solver) and RBFOpt results for all Process conditions. The cost columns report median reconstructed search-completion costs over all ten seeds; hit counts concern the frozen numerical oracle.

		RBFOpt (solver)		RBFOpt	
Case	B	Target hits	Cost (s)	Target hits	Cost (s)
Column	16	4/10	37.35	5/10	33.54
CSTR 5	8	10/10	11.01	10/10	10.45
CSTR 10	15	10/10	22.17	9/10	20.09
CSTR 15	22	10/10	35.02	8/10	31.85
CSTR 20	29	9/10	52.57	9/10	44.47
CSTR 25	36	10/10	72.73	10/10	56.86
CSTR 30	43	10/10	64.27	10/10	38.89
Small Batch	10	5/10	6.02	5/10	5.12
Dynamic	12	9/10	13.28	10/10	12.29
Dynamic	15	10/10	18.08	10/10	16.10
Dynamic	20	10/10	25.45	10/10	20.54
Dynamic	30	10/10	51.09	10/10	34.19
Dynamic	37	10/10	78.41	10/10	42.42
Dynamic	50	10/10	164.01	10/10	65.14

S10.2 Paired differences and sampling controls

Paired success differences retain the common case, budget and seed. For Grid, the descriptive percentile bootstrap resamples the 21 intact function vectors, each containing ten paired seed differences, rather than treating 210 observations as exchangeable Bernoulli draws. For Process, paired seeds are resampled within each of the 14 conditions; the six related Dynamic budgets are not pooled for inference. The seed is 20261020 with 2,000 bootstrap replications and deterministic case/seed ordering. Intervals are exploratory and unadjusted for multiple comparisons; ten-seed boundary cases may yield degenerate intervals, and zero inclusion does not establish equivalence. Full condition-level comparisons are in `manuscript/tables/paired_comparisons.csv`, with the independently preserved FMQA–uniform analysis under `results/review_comparison_20261020/fmqa_uniform/`.

Table S18: Selected paired success differences, in percentage points. The first method minus the second defines the sign; Process intervals concern the stated condition only.

Grid

B	Contrast	Difference	95% interval
10	RBFOpt – RBFOpt (solver)	-1.43	[-6.67, 3.81]
100	RBFOpt – RBFOpt (solver)	-0.48	[-6.19, 5.71]
10	RBFOpt – Uniform	13.33	[5.71, 24.29]
100	RBFOpt – Uniform	60.00	[42.86, 75.71]
10	BAM – BAM-prefix control	0.00	[0.00, 0.00]
100	BAM – BAM-prefix control	10.95	[0.48, 23.81]
10	FMQA-SA – Uniform	0.95	[-0.95, 2.86]
100	FMQA-SA – Uniform	40.00	[28.57, 52.38]
10	DSDA – RBFOpt	-13.81	[-24.29, -6.19]
100	DSDA – RBFOpt	-31.90	[-46.19, -18.10]

Process: ten paired seeds per condition

Case	B	Contrast	Outcomes	Difference	95% interval
Dynamic	12	RBFOpt – DSDA	4/6/0/0	60	[30, 90]
Small Batch	10	FMQA-SA – Uniform	2/4/2/2	20	[-30, 60]

Outcomes list both successful / first only / second only / neither successful. The descriptive intervals resample functions for Grid and paired seeds within each Process condition; they do not establish general superiority or equivalence.

S10.3 Fixed-path point-cost scenarios and suite exclusions

For each retained trajectory we evaluate the following fixed-path cost scenario.

$$\widehat{T}_\alpha = T_{\text{search}} + \alpha \sum_q c(\xi^{(q)}) + t_{\text{prep}}, \qquad \alpha \in \{0.01, 0.1, 1, 10, 100\}.$$

This changes saved point costs only, retaining trajectories, target indicators and one-time preparation; it does not predict adaptation under new wall-clock limits. The relative importance of the candidate-search overhead decreases as evaluation costs dominate. Excluding Xin-She Yang 3 and Bohachevsky 2 preserves the main descriptive pattern. For the sampled Bohachevsky 2 lattice, $\cos(3\pi a_i) = 1$ at every coordinate, leaving an objective nearly redundant with sampled Sum of Squares after normalization. Without both functions, B=100 rates are 79.47% for RBFOpt, 80.00%

for RBFOpt (solver), 66.84% for BAM, 57.89% for FMQA-SA, 54.74% for the BAM-prefix control, 44.21% for DSDA, and 17.89% for uniform. At B=10, BAM and its prefix control each retain 47.89%, compared with RBFOpt 15.79% and RBFOpt (solver) 16.84%. The full exclusion tables remain in `results/review_comparison_20261020/tables/grid_exclusions.csv`.

Table S19: Median Grid B=100 completion costs in fixed-path point-cost scenarios, in seconds. The original recorded trajectories are retained.

α	DSDA	RBFOpt	BAM	FMQA-SA	Uniform	RBFOpt (solver)
0.01	0.24	8.15	1.06	62.69	0.61	307.42
0.1	2.42	14.85	6.89	68.13	6.05	311.40
1	24.14	70.81	64.42	124.13	60.51	365.74
10	241.41	619.68	640.12	685.25	605.06	1015.89
100	2414.06	6101.70	6397.08	6295.95	6050.62	6343.39

S10.4 Limited reconstructed FMQA diagnostics

A separate implementation audit and bounded diagnostic analysis retain the original benchmark paths and performance timings (SECQUOIA, 2026d). Three runs were fixed before fitting: Rosenbrock B10 seed0, CSTR15 B22 seed0, and Dynamic B12 seed0. The selection rule takes the first SA-selected proposal, the first uniform-fallback proposal when present, and the final proposal, removing duplicates. This yields seven distinct iterations: Rosenbrock 1 and 7, CSTR15 1 and 19, and Dynamic 1, 2 and 9. The selected archives lack the original fitted weights and QUBOs, so the seven models were reconstructed from pre-proposal observations, recorded targets, seeds, and archived sources without new SA calls or benchmark runs. All 3,328 enumerated binary strings satisfy the recorded combined absolute/relative FM–QUBO–BQM energy-identity tolerance, with absolute tolerance 10^{-6} and relative tolerance 10^{-5}. Stored sample energies agree with reconstruction under absolute and relative tolerances of 10^{-5}; their largest absolute difference is approximately 1.328×10^{-6}. This agreement does not establish identity of the historical and reconstructed energy landscapes at unsampled states. The five selected SA proposals attain the eligible minimum of their respective reconstructed models. The two selected Dynamic fallback iterations have no eligible stored SA sample and an out-of-domain binary minimum in the reconstructed model. Their selected-point gaps above the reconstructed eligible minimum are approximately 2.178316 and 0.384912 in scaled-energy units. These purposively selected examples illustrate admissibility and fallback behavior without estimating prevalence or isolating causal contributions of learning, annealing and fallback. Synthetic constant-model checks show that zero coefficient scaling produces nonfinite coefficients and a propagated wrapper error before annealing or uniform fallback. The occurrence of that boundary in historical fitted models remains unknown. Input hashes, reconstructed fitted weights, training prefixes, exhaustive energies, and recorded-sample comparisons are preserved under `results/review_diagnostics_20261020/fmqa/`.

References

Clopper, C.J., Pearson, E.S., 1934. The use of confidence or fiducial limits illustrated in the case of the binomial. Biometrika 26, 404–413. doi:10.1093/biomet/26.4.404.

D-Wave Systems, n.d. dwave-samplers: Classical algorithms for solving binary quadratic models. URL: https://github.com/dwavesystems/dwave-samplers/releases/tag/1.6.0. Version 1.6.0; classical CPU simulated annealing; accessed October 15, 2026.

Nannicini, G., 2021. On the implementation of a global optimization method for mixed-variable problems. Open J. Math. Optim. 2, 1–25. doi:10.5802/ojmo.3.

SECQUOIA, 2026a. Black-Box-for-GDP: additional genetic-search and sampling comparisons. URL: https://github.com/SECQUOIA/Black-Box-for-GDP/tree/6e083eb49b497ce4fea90131e18cc0459db682e8/results/review_comparison_20261020. Analysis snapshot on review/additional-comparisons, accessed October 20, 2026; original execution provider 38d917ef75ef6767c2a9f238d88be19bf0b89160; private pending administrator-enabled public release.

SECQUOIA, 2026b. Black-Box-for-GDP: analytical Grid-GDP and process benchmark archive. URL: https://github.com/SECQUOIA/Black-Box-for-GDP/tree/cb70771b2037301bb6d0e38a2362d49b53fcda7d. Snapshot of main; commit cb70771b2037301bb6d0e38a2362d49b53fcda7d, accessed October 19, 2026.

SECQUOIA, 2026c. Black-Box-for-GDP: CSTR15 fixed-Boolean identity preprocessing and tightened-feasibility validation. URL: https://github.com/SECQUOIA/Black-Box-for-GDP/tree/854a1a7fc5a20a5489d82ef7991a16aaf89025f3/results/review_feasibility_tautology_20261020. Additional numerical validation of the archived W03 configuration, accessed October 20, 2026; original benchmark records and execution revisions preserved; private pending public release.

SECQUOIA, 2026d. Black-Box-for-GDP: limited FMQA and numerical-feasibility diagnostics. URL: https://github.com/SECQUOIA/Black-Box-for-GDP/tree/15663134704b7bb1d4804c94c70edabaffbf8ecd/results/review_diagnostics_20261020. Diagnostic snapshot accessed October 20, 2026; reconstructed fits are distinct from unsaved historical weights; earlier snapshot preceding the separate CSTR15 tightened-feasibility follow-up.

SECQUOIA, n.d. Black-Box-for-GDP: historical QHDOpt GDP formulation. URL: https://github.com/SECQUOIA/Black-Box-for-GDP/blob/11ba8728a107d6d16ed63e72d334a66662f9ebc1/qhdopt_gdp.py. Historical source; commit 11ba8728a107d6d16ed63e72d334a66662f9ebc1; accessed October 19, 2026.

Wu Group, n.d. Quantum Hamiltonian descent: 2D test functions. https://jiaqileng.github.io/quantum-hamiltonian-descent/nonconvex2d.html. Source collection for the 21 adapted finite grids used here; accessed September 29, 2026.